

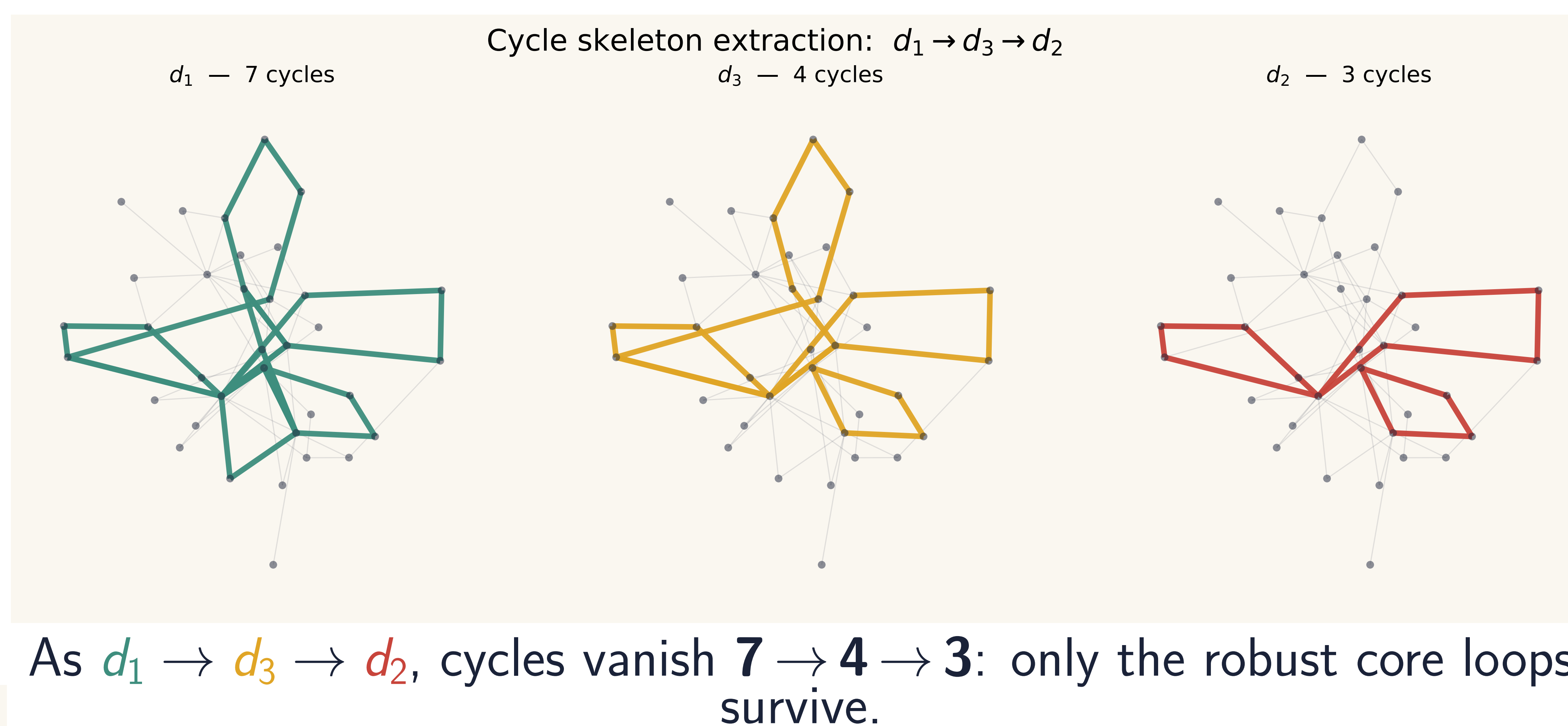


The big idea

Distance is a **tunable resolution knob** on musical structure. Sweeping $d_1 \rightarrow d_3 \rightarrow d_2$ distills a piece down to its **core cyclic skeleton**.

- Three path distances: d_1 (min edges), d_2 (min weight, the only metric), d_3 (hybrid).
- Proven $d_2 \leq d_3 \leq d_1$ and barcode injection $B_2 \subseteq B_3 \subseteq B_1$.
- Validated on real Korean court music — an exact match to the paper.

Cycle skeleton extraction (Sanghyeondodeuri, geomungo)



From music to a graph

Each **node** = (pitch, duration); an **edge** joins consecutive notes; the **weight** is $1/(\text{co-occurrence count})$ — frequent neighbors are closer. Rests are skipped.

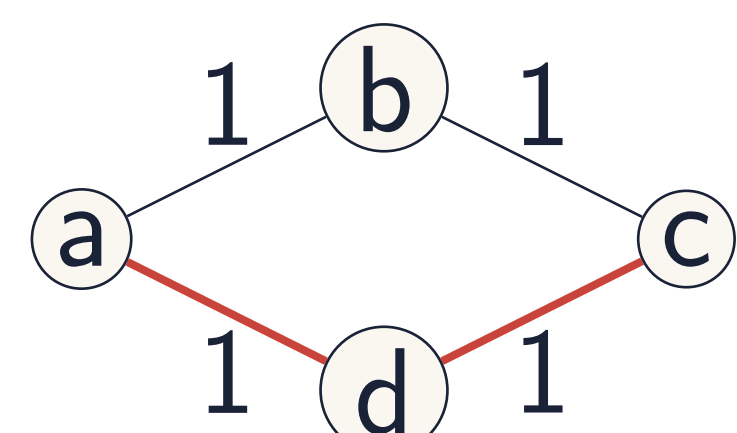
Three path distances

For nodes v, w on the music graph $M = (V, E, W)$:

d_1 **min-edge path** — fewest edges (weighted sum along it).

d_2 **min-weight path** — least total weight; *the only true metric*.

d_3 **hybrid** — min weight *among* the fewest-edge paths.



Path-based distance uses existing edges (red), not a new direct link.

Only d_2 is a metric

d_2 satisfies the triangle inequality; d_1 and d_3 can violate it (a heavy direct edge). Yet all three give meaningful musical structure.

Main result: a universal ordering

For every pair of nodes,

$$d_2(v, w) \leq d_3(v, w) \leq d_1(v, w).$$

This carries over to one-dimensional persistent homology.

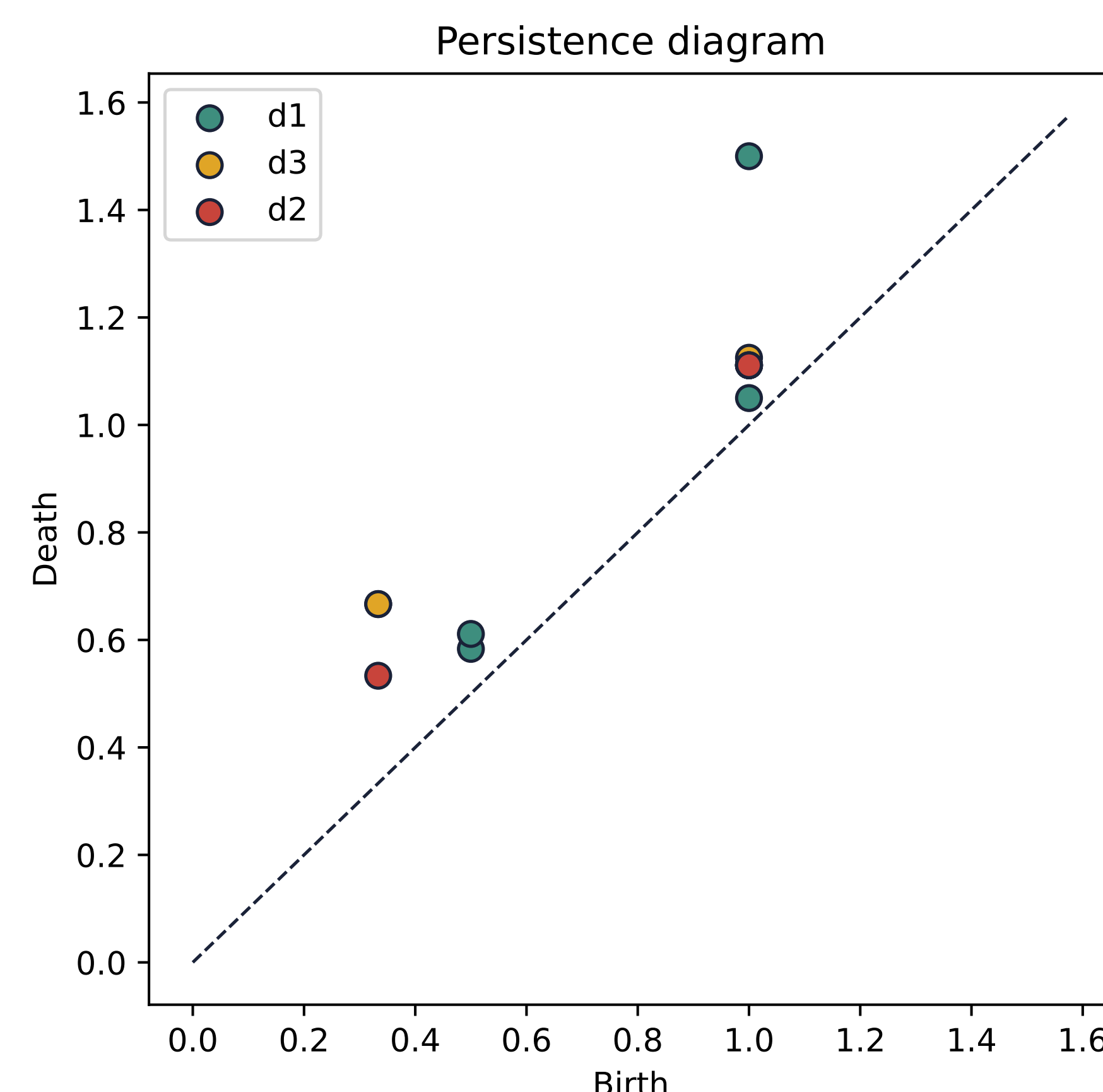
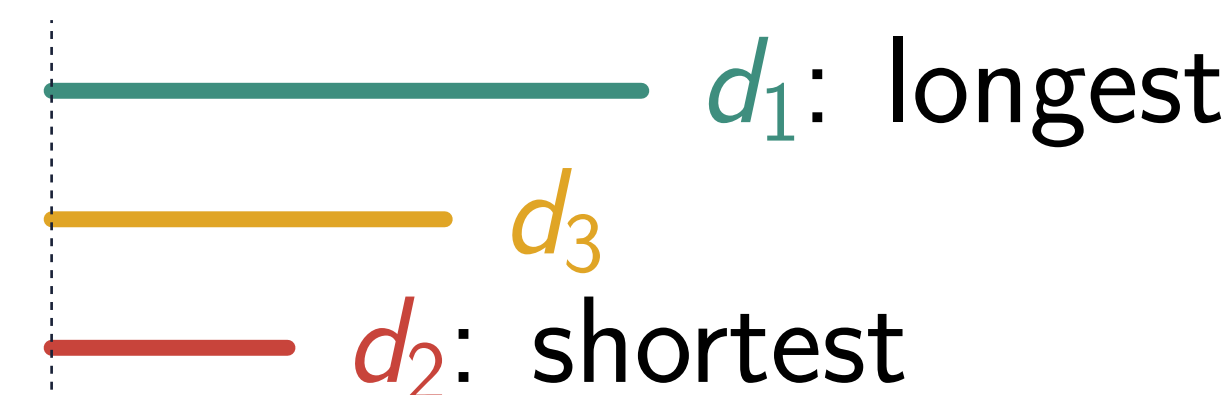
Barcode injection $B_2 \subseteq B_3 \subseteq B_1$

The birth edges nest, $B_2 \subseteq B_3 \subseteq B_1$, giving injections

$$\text{bcd}_1(d_2) \hookrightarrow \text{bcd}_1(d_3) \hookrightarrow \text{bcd}_1(d_1).$$

Each cycle keeps its **birth**; its **lifespan grows** $d_2 \leq d_3 \leq d_1$, and the number of cycles can only increase.

shared birth

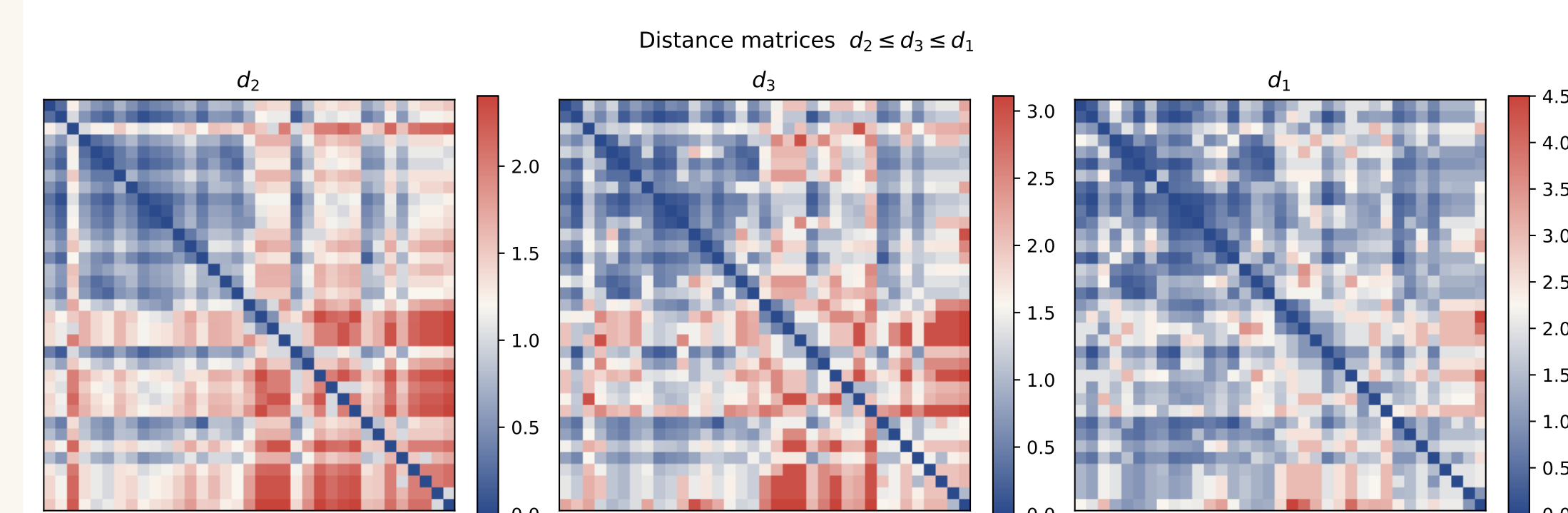


On real music: d_1 points lie above d_2 — longer lifespans, shared births.

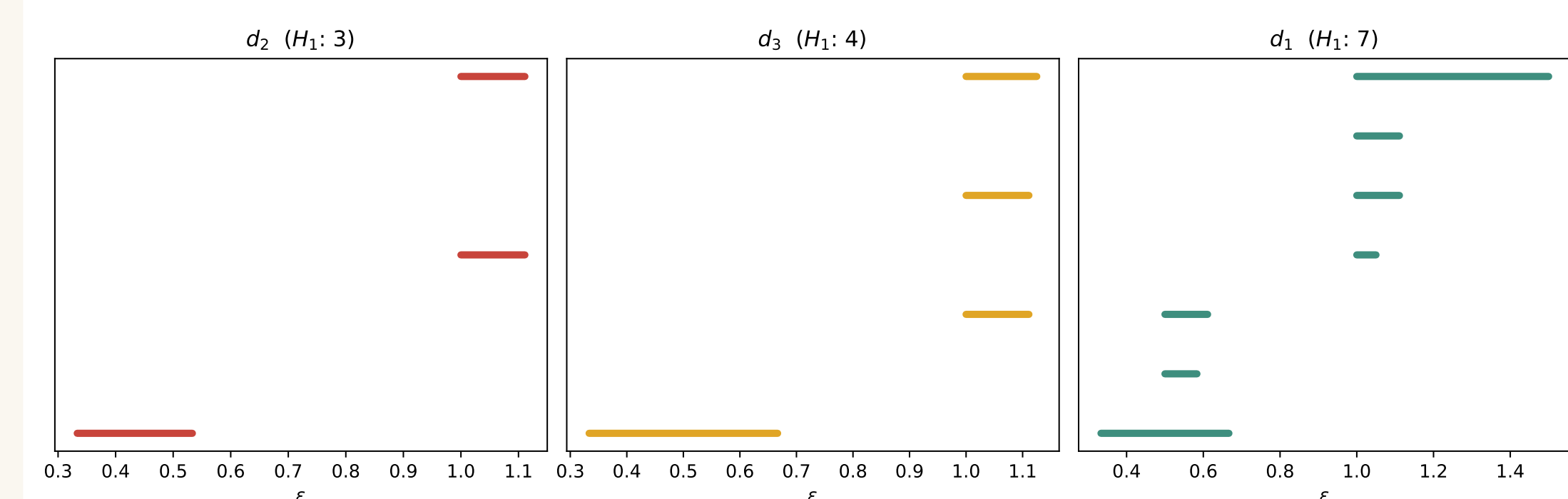
Why it matters

The same piece yields *different but nested* topological readings. d_2 exposes the robust core; d_1 adds finer, peripheral loops. The changes are **systematic, not random**.

Evidence on real music



Distance matrices: d_2 (cool/smooth) $\rightarrow d_1$ (hot) — the ordering is visible.



H_1 barcodes: **7 / 4 / 3** bars for $d_1/d_3/d_2$; lifespans grow, births shared.

Application: distilling the core

Ranking d_1 cycles by how far down they survive ($\rightarrow d_3 \rightarrow d_2$) gives a principled **importance hierarchy of musical loops** — a multi-resolution descriptor of a piece.

Takeaways & future work

- Distance choice = a tunable lens; results are nested, not arbitrary.
- Toward **AI music composition** with multiple stylistic perspectives from one source.



Interactive explorer & audio demos on the project page.

